

1. A Hospital wants to estimate the average length of stay μ (in days). It takes a random sample of $n = 100$ patients. From the sample it calculates $\bar{x} = 4.53$ and $s = 3.68$

- (a) Calculate a 95% confidence interval for μ .

- (b) Calculate a 99% confidence interval for μ .

- (c) Calculate a 90% confidence interval for μ .

2. The hospital made a typo in reporting its data. The random sample actually only had $n = 10$ patients.
- (a) Calculate a 95% confidence interval for μ .

(b) Calculate a 99% confidence interval for μ .

(c) Calculate a 90% confidence interval for μ .

3. You want to estimate the proportion (p) of Trinity students that are left-handed. You sample $n = 100$ random students and find $\hat{p} = \frac{17}{100}$.

(a) Is $\hat{p}$ a binomial random variable?

(b) Calculate a 95% confidence interval for p .

(c) Calculate a 99% confidence interval for p .

(d) Calculate a 90% confidence interval for p .

